

OE-EC803A — Internet of Things

Semester VIII • 3L:0T:0P • 3 Credits

Module 3: Prototyping: Thinking & Embedded Hardware Platforms

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Thinking About Prototyping

Prototyping is the iterative process of quickly assembling a working model of a device to test assumptions and functionality before entering mass manufacturing.

1.1 Prototyping Considerations

- **Sketching:** Fast, non-functional drawings to visualize hardware forms and user flows.
- **Familiarity:** Choosing tools the designer already knows (e.g. Arduino IDE) to reduce development time, even if other tools are mathematically more optimal.
- **Costs vs. Ease:** Fast prototyping tools (like jumper wires and breadboards) are cheap and easy, but produce bulky, fragile systems. Production-ready designs require custom PCBs.
- **Open Source vs. Closed Source:** Open hardware/software allows developers to adapt existing code libraries and hardware schematics, tapping into massive global communities.

2. Basic Electronics for IoT

IoT edge nodes interface directly with the physical world using sensors and actuators.

2.1 Basic Quantities

Ohm's Law and Electrical Power

Ohm's Law relates voltage (V), current (I), and resistance (R):

$$V = I \cdot R \quad (1)$$

Electrical power (P) in watts is given by:

$$P = V \cdot I = I^2 \cdot R \quad (2)$$

Voltage Dividers and Analog-to-Digital Conversion (ADC)

Voltage Divider Formula: Reduces a high analog voltage (V_{in}) to a safe range (V_{out}) for microcontroller pins:

$$V_{out} = V_{in} \cdot \left(\frac{R_2}{R_1 + R_2} \right) \quad (3)$$

ADC Resolution: Quantizes an analog voltage into a digital integer. For an n -bit ADC with reference voltage V_{ref} , the step size (Resolution) is:

$$\text{Step Size (LSB)} = \frac{V_{ref}}{2^n} \quad (4)$$

The measured digital value D for an input voltage V_{in} is:

$$D = \text{round} \left(\frac{V_{in}}{\text{Step Size}} \right) = \text{round} \left(V_{in} \cdot \frac{2^n}{V_{ref}} \right) \quad (5)$$

2.2 Signals and Interface Elements

- **Digital vs. Analog:** Digital signals are binary (0 or 1, representing 0V and 3.3V/5V). Analog signals vary continuously (e.g., LDR resistance changing with light).
- **PWM (Pulse Width Modulation):** Mimics analog outputs from a digital pin by switching the pin ON and OFF very rapidly. The ratio of ON time to the period is called the **Duty Cycle** (D):

$$V_{\text{eff}} = D \cdot V_{\text{max}} \quad (6)$$

- **DHT11 Sensor:** A digital sensor measuring temperature and humidity, sending data over a single-wire digital protocol.
- **Relay:** An electromagnetic switch allowing a low-power microcontroller pin to switch high-voltage/high-current AC devices.

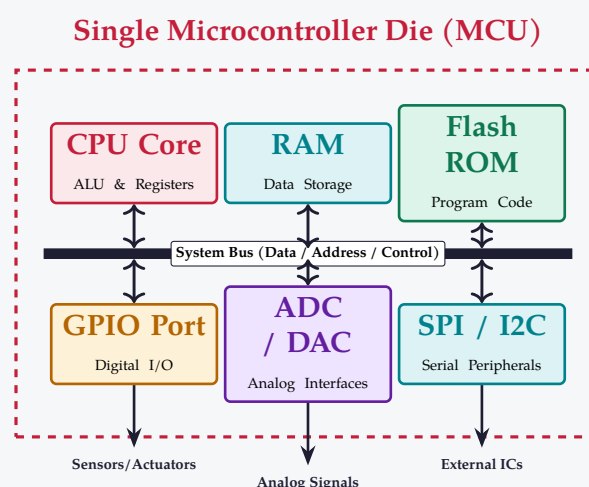
3. Prototyping Embedded Platforms

IoT designs select hardware platforms based on processing power, OS requirements, and connectivity.

3.1 Embedded Computing Basics

- **Microcontroller (MCU):** A single integrated circuit containing a CPU, RAM, Flash memory, and I/O peripherals (GPIO, ADC, SPI, I2C). Low-power, real-time execution. Example: ATmega328P.
- **Microprocessor (MPU):** Contains only the CPU. Requires external RAM, Flash, and support ICs. Runs complex Operating Systems (Linux). High power consumption. Example: ARM Cortex-A72.

Microcontroller Internal Architecture (System Bus Model)



3.2 Hardware Platforms Overview

1. Arduino (Microcontroller-Based):

- Powered by the **ATmega328P** 8-bit MCU operating at 16 MHz, containing 32 KB Flash, 2 KB SRAM, and 1 KB EEPROM.
- Programmed in C++ using bare-metal register manipulation (e.g. setting `DDRB |= (1 << 5)` to configure Pin 13 as output, and writing to `PORTB` to toggle pins) compiled via `avr-gcc`.
- Ideal for low-power sleep modes ($< 15 \mu\text{A}$ in power-down mode) and deterministic, timing-critical sensor polling.

2. Raspberry Pi (Single-Board Computer):

- Powered by a Broadcom **System-on-Chip (SoC)** containing a quad-core 64-bit ARM Cortex processor, up to 8 GB RAM, and GPU.
- Runs a fully pre-emptive Linux OS (Raspberry Pi OS) from an SD card. This introduces OS scheduling latency (non-deterministic) but allows full network stacks, file systems, and high-level runtimes (Python, Node.js).
- Used as local edge gateways, database hosts, and for CPU-intensive tasks like computer vision or edge AI inference.

3. BeagleBone Black (PRU Co-processors):

- Features a 1 GHz ARM Cortex-A8 processor running Linux, augmented with two auxiliary **PRUs (Programmable Real-Time Units)**.
- Each PRU is a 32-bit RISC core operating at 200 MHz with direct access to hardware I/O pins and shared RAM, operating independently of the Linux kernel's interrupts or process schedules.
- Solves the "non-deterministic OS" issue, enabling sub-microsecond GPIO toggling, precise high-speed industrial motor control, and custom hardware protocol decoding.

4. Electric Imp (Cloud-Integrated Module):

- A secure hardware module featuring an ARM Cortex processor and Wi-Fi interface.
- Utilizes a **Cloud-Device Split Model**: Code is split into a local *Device* script (interfacing with pins) and a cloud-based *Agent* script (running in Electric Imp's secure VM, handling HTTPS/OAuth).
- Programmed in the **Squirrel** scripting language. Features built-in secure boot, automated secure OTA firmware updates, and managed cloud routing.

5. Mobile Phone and Tablets (Sensors & Application Runtimes):

- High-performance mobile devices equipped with gigabytes of RAM, multi-core CPUs, and multi-Gbps cellular/Wi-Fi/Bluetooth networking.
- Host rich integrated sensors (dual-frequency GPS, 3-axis accelerometers, gyroscopes, magnetometers, ambient light sensors, cameras, and microphones).
- Used for rapid application-level prototyping, mobile dashboard display, and utilizing internal sensors as edge input devices.

6. Plug Computing (Always-On Mini-Servers):

- Compact, wall-plug-sized, always-on servers containing a low-power MPU (e.g. Marvell Kirkwood ARM), RAM, flash storage, and an Ethernet/Wi-Fi connection.
- Run headless Linux server distributions (e.g., Debian).

- Act as quiet, unobtrusive, low-power local gateways, print servers, and local home automation hub controllers.

Quick Revision — Module 3

Embedded Hardware Platforms | OE-EC803A

Key Formulas and Metrics at a Glance

Ohm's Law	$V = I \cdot R$ (Volts, Amperes, Ohms)
Power dissipation	$P = V \cdot I = I^2 \cdot R$ (Watts)
PWM Effective Voltage	$V_{\text{eff}} = D \cdot V_{\text{max}}$ ($D = \text{Duty Cycle} \in [0, 1]$)
ADC Resolution	$V_{\text{step}} = \frac{V_{\text{ref}}}{2^N - 1}$ (N -bit Analog-to-Digital Converter)

One-Line Definitions

- **Microcontroller:** An IC integrating CPU, memory, and peripheral interfaces on a single chip.
- **Single-Board Computer:** A complete computer built on a single circuit board, running an operating system.
- **PRU (BeagleBone):** Co-processors that run independently of the main OS to execute real-time I/O tasks.
- **PWM:** A technique to simulate analog voltages by pulsing a digital output at varying widths.
- **Relay:** An electromechanical switch used to control high-power circuits with low-power digital signals.

Frequently Asked / Viva Questions

Q1. What is the difference between a microcontroller and a microprocessor?

A microcontroller integrates the CPU, RAM, Flash memory, and I/O peripherals on a single silicon chip (System-on-Chip). A microprocessor contains only the CPU core and requires external components (RAM, Storage, Ethernet ICs) to function.

Q2. Why are microcontrollers preferred over microprocessors for simple IoT sensor nodes?

Microcontrollers consume far less power (milliwatts vs. watts), support deep sleep modes that allow battery operation for years, are much cheaper, and run code directly on bare-metal without the boot latency of an operating system.

Q3. What are the PRUs in a BeagleBone Black, and why are they useful?

The Programmable Real-Time Units (PRUs) are two auxiliary 32-bit RISC cores running at 200MHz independently of the main Linux OS. They allow BBB to

Frequently Asked / Viva Questions (contd.)

perform fast, deterministic real-time I/O tasks without scheduling delays caused by the Linux kernel.

Q4. Explain the concept of PWM and how it is used to dim an LED.

PWM (Pulse Width Modulation) switches a digital pin between high (5V) and low (0V) states rapidly. By changing the Duty Cycle (the fraction of time the pin is high), the effective average voltage varies continuously. This average voltage controls the LED brightness.

Q5. What is a Single-Board Computer (SBC)? Give an example.

A Single-Board Computer is a complete, functioning computer built on a single circuit board, incorporating a microprocessor, memory, networking, and USB ports. An example is the Raspberry Pi, which runs a Debian-based Linux OS.

Q6. How does a relay isolate a microcontroller from high-voltage appliances?

A relay uses an electromagnet. When the microcontroller energizes the internal coil (isolated low-voltage side), the magnetic field pulls a mechanical contact switch closed on the high-voltage side. There is no electrical connection between the two sides.

Q7. Why is community support considered an important parameter when choosing prototyping hardware?

IoT development moves fast. A large active community ensures open-source driver libraries are available for almost any sensor, troubleshooting forum threads exist for common bugs, and development boards are widely stocked.

Q8. What is Electric Imp, and how does it handle programming?

Electric Imp is a secure IoT module integrating Wi-Fi hardware with a managed cloud. It is programmed in the Squirrel scripting language, allowing developers to write both local device code and cloud agent code in a single framework.

Comparison Snapshot

Arduino vs. Raspberry Pi vs. BeagleBone Black:

Feature	Arduino Uno	Raspberry Pi 4	BeagleBone Black
Type	Microcontroller (MCU)	Single-Board Computer (SBC)	Industrial SBC
Processor	8-bit ATmega328P (16MHz)	64-bit Quad ARM (1.5GHz)	32-bit ARM (1GHz) + 2x PRUs
Memory	2 KB SRAM, 32 KB Flash	2GB / 4GB / 8GB LPDDR4	512MB DDR3, 4GB eMMC
Operating System	None (Bare-Metal)	Yes (Linux / Raspberry Pi OS)	Yes (Debian Linux)
Power Draw	Very Low (50mW)	Medium-High (3W to 5W)	Medium (1.5W to 2.5W)
Best Use Case	Analog sensor read, relays.	Computer vision, IoT gateway.	Real-time industrial automation.